

Airline Customer Recommendation Prediction Using CRISP-DM and Machine Learning

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1.Introduction and Background:

The airline industry operates in a highly competitive and reputation-sensitive environment where customer perception directly influences revenue, brand positioning, and long-term sustainability. As airlines offer a largely standardised core service (transportation), differentiation is primarily driven by service quality, customer experience, and perceived reliability.

Customer recommendations function as a form of electronic word-of-mouth (eWOM), where passengers publicly share experiences through review platforms and booking websites. Research shows that peer-generated content is often more influential than corporate messaging, particularly in high-involvement services such as air travel. As a result, recommendation behaviour serves as a strong proxy for customer satisfaction and loyalty.

Customer satisfaction in airlines is commonly explained using frameworks such as SERVQUAL, which evaluates service quality across dimensions including reliability, responsiveness, assurance, empathy, and tangibles. Among these, frontline employee behaviour plays a critical role, where interpersonal factors such as courtesy and perceived friendliness can significantly shape overall passenger experience.

However, modern airline feedback is increasingly unstructured, with large volumes of textual data such as reviews and complaints complementing traditional rating systems. While numerical ratings provide structured signals, they often fail to capture emotional nuance, sarcasm, or contextual dissatisfaction, creating a gap between measured satisfaction and actual sentiment.

This has driven the adoption of predictive analytics and machine learning in airline customer experience management. Techniques such as natural language processing (NLP), sentiment analysis, and classification models enable extraction of patterns from both structured and unstructured data, supporting prediction of customer recommendation behaviour and identification of service issues.

Despite this progress, integrating numerical and textual data remains challenging, as they may present conflicting signals—for example, high ratings alongside negative written feedback. This highlights the importance of combined analytical approaches to fully understand customer perception.

Overall, this study sits at the intersection of service quality theory and machine learning, aiming to transform fragmented airline customer feedback into actionable insights that support operational improvement and enhanced customer experience.

2. Methodology

This study adopts the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework to structure the analytical workflow for predicting airline customer recommendation behaviour using numerical and textual review data.

Business Understanding

The objective is to predict whether passengers recommend an airline based on service quality and review behaviour. In the airline industry, recommendation is a key indicator of satisfaction, loyalty, and perceived service quality in a highly competitive environment (Wirtz & Lovelock, 2016). The goal is to identify the most influential drivers of customer recommendation to support managerial decision-making.

Data Understanding

The dataset includes numerical service ratings, categorical travel variables, and unstructured textual reviews. This mixed structure increases analytical complexity, as satisfaction is shaped by both measurable service attributes and subjective feedback (Liu, 2012).

Initial exploration revealed significant missing values, particularly in Wifi & Connectivity and Inflight Entertainment. Inconsistencies were also observed between Overall Rating and recommendation behaviour, where some high-rated customers did not recommend the airline.

Data Preparation

Preprocessing was performed to improve data quality and reduce bias from missing values. Date variables were standardised into datetime format for consistency.

Numerical rating variables were imputed using a group-based median strategy based on Seat Type, allowing context-aware estimation. This was compared with global median imputation before final selection.

Categorical missing values were handled by introducing an “Unknown” category to retain observations. Variables with excessive missingness (e.g., Aircraft) were removed. Missingness indicator variables were also created for features with systematic absence to preserve potential behavioural signals.

Text data was cleaned using tokenisation and stopword removal to extract key themes from reviews. While this improves structure for analysis, it may reduce sensitivity to sarcasm and nuanced emotional expressions common in airline feedback.

Descriptive Statistics and Visualisation

Descriptive statistics (mean, median, standard deviation, and distribution analysis) were used to examine service rating patterns.

Visualisations were selected based on data type and analytical purpose. Histograms and density plots assessed distribution changes after imputation. Boxplots and violin plots compared service ratings across recommendation groups. Correlation heatmaps identified relationships between service attributes and recommendation behaviour. Additional managerial visualisations highlighted airline-level performance differences and rating–recommendation contradictions.

Modelling Approach

Two supervised learning models were used: Logistic Regression and Random Forest.

Logistic Regression was chosen for binary classification due to its interpretability in explaining relationships between service features and recommendation outcomes (Hosmer et al., 2013), although it assumes linear relationships.

Random Forest was used as a complementary ensemble method to capture non-linear interactions between variables (Breiman, 2001). Hyperparameters such as tree depth and number of estimators were tuned to improve stability and performance.

Evaluation Strategy

The dataset was split into training and testing sets to evaluate generalisability. Performance was measured using accuracy, precision, recall, and F1-score to ensure balanced evaluation across classes.

A robustness test was also conducted by training on one subset of airlines and testing on unseen airlines to assess model stability across different operational contexts.

Overall Summary

The CRISP-DM framework ensured a structured workflow connecting business understanding, data preparation, modelling, and evaluation into a coherent analytical process.

3. Data Pre-Processing (Data Cleaning and Preparation)

The data preparation stage focused on improving data quality, reducing analytical bias, and ensuring suitability for predictive modelling of airline customer recommendation behaviour. Particular attention was given to missing data patterns, inconsistencies between customer ratings and recommendation behaviour, and the preparation of textual review content for analysis.

Numerical Data Cleaning

Several non-trivial data quality issues were identified during exploratory analysis. First, inconsistencies were observed between Overall Rating and customer recommendation behaviour, where some passengers provided high overall ratings while still not recommending the airline. This suggests that recommendation behaviour may reflect broader service expectations beyond numerical satisfaction scores alone, consistent with prior customer satisfaction research (Anderson & Sullivan, 1993).

Second, substantial missingness was identified across several service-quality variables, particularly Wifi & Connectivity, Inflight Entertainment, and Aircraft. The missing patterns appeared systematic rather than completely random, as certain services were more frequently absent for specific seat types or travel contexts.

To address missing data, two alternative imputation strategies were evaluated. The first strategy used global median imputation, which is computationally simple but may distort relationships between passenger groups by ignoring service context differences. The second strategy applied group-based median imputation using Seat Type categories to estimate missing service ratings within similar passenger segments. This approach was selected because it preserves contextual differences between travel classes while remaining robust to skewed rating distributions and outliers commonly observed in customer review datasets.

A comparison between global imputation and group-based imputation was conducted using distribution analysis of Wifi & Connectivity ratings. The group-based approach retained more realistic rating distributions across passenger categories and reduced the risk of oversmoothing the data. Additional missingness indicator variables were also created for highly incomplete variables to preserve potential behavioural information associated with non-response patterns.

Categorical variables with missing values were assigned an “Unknown” category to retain observations without introducing artificial assumptions. Variables with extremely high

missingness, such as Aircraft, were removed from the modelling process due to limited analytical reliability.

Textual Data Preprocessing

A preprocessing pipeline was developed for customer review text to support exploratory text analytics. Reviews were converted to lowercase and processed using token extraction and stopwords removal techniques to identify meaningful service-related patterns within customer feedback. Common filler words were removed to reduce noise while preserving operationally relevant terms associated with airline service experiences.

The preprocessing approach aimed to identify recurring patterns within positive and negative reviews rather than performing advanced sentiment classification. Word-frequency analysis revealed recurring negative themes related to delays, poor service experience, and customer dissatisfaction, while positive reviews frequently referenced staff helpfulness, comfort, and service quality.

However, several limitations remain within this preprocessing strategy. Basic word-frequency approaches struggle to capture contextual meaning, sarcasm, negation, and emotional nuance within airline reviews. For example, phrases containing implicit dissatisfaction or contradictory sentiment may not be accurately represented through simple token frequency analysis. Additionally, removal of certain contextual words may reduce semantic richness and weaken interpretation of customer emotions.

Overall, the preprocessing stage improved dataset consistency and analytical usability while acknowledging the trade-offs between data simplification, contextual accuracy, and predictive interpretability.

4. Data Analytics

4.a Descriptive and Predictive Analytics (Numerical Features)

Descriptive analysis of numerical service-rating variables revealed several important patterns associated with customer recommendation behaviour. Summary statistics showed generally low-to-moderate average ratings across service attributes, particularly for Wifi & Connectivity and Inflight Entertainment, indicating widespread dissatisfaction in selected operational service areas. In contrast, Cabin Staff Service received comparatively higher average ratings,

suggesting that human interaction quality remained stronger than technology-related service components.

A non-obvious pattern emerged when comparing Overall Rating and recommendation behaviour. Although high ratings were generally associated with positive recommendations, a subset of customers provided high overall ratings while still not recommending the airline. This suggests that recommendation behaviour is influenced not only by satisfaction scores but also by broader service expectations and perceived reliability, consistent with customer loyalty research (Jones & Sasser, 1995).

Correlation analysis revealed that Value For Money had one of the strongest relationships with customer recommendation behaviour, indicating that perceived economic value may influence recommendation decisions more strongly than individual service attributes alone. Additionally, several service-quality variables showed moderate intercorrelation, suggesting overlapping dimensions of customer experience.

Five key visualisations were developed to support interpretation of the numerical analysis:

1. Distribution histograms showing the spread and skewness of service-rating variables.
2. Boxplots comparing Cabin Staff Service ratings across recommendation outcomes.
3. Violin plots examining differences in Value For Money ratings between recommended and non-recommended airlines.
4. Correlation heatmaps illustrating relationships between service-quality variables and recommendation behaviour.
5. Airline-level comparison charts highlighting differences in overall ratings and perceived value across major airlines.

In addition, a contradiction-focused visualisation was developed for management interpretation, illustrating customers who provided high overall ratings but still did not recommend the airline. This visualisation was designed to support managerial understanding of gaps between satisfaction and customer advocacy.

For predictive analytics, the dataset was divided into training and testing subsets using an 80:20 split. Two supervised machine learning models were trained: Logistic Regression and Random Forest Classification.

Logistic Regression was selected due to its interpretability and suitability for binary classification problems. Random Forest was selected because of its ability to capture non-linear relationships and interaction effects between service-quality variables.

Hyperparameter tuning was conducted for the Random Forest model by adjusting the number of estimators, maximum tree depth, and minimum sample split thresholds to improve predictive stability and reduce overfitting.

A robustness test was also conducted by training the model on one subset of airlines and testing it on different airlines. The results showed that predictive performance remained relatively stable across airline groups, although slight performance variation suggested that customer experience patterns differ across carriers.

Model evaluation was conducted using accuracy, precision, recall, and F1-score. Random Forest slightly outperformed Logistic Regression across most evaluation metrics, achieving stronger predictive performance while maintaining classification stability. However, Logistic Regression provided greater interpretability for managerial explanation of service-related influences on recommendation behaviour.

4.b Descriptive and Predictive Analytics

Textual review analysis was conducted to identify recurring themes and linguistic patterns associated with customer recommendation behaviour. Reviews were separated into positive and negative recommendation groups, and word-frequency analysis was performed after preprocessing and stopword removal.

The analysis revealed clear differences between positive and negative review language. Positive reviews frequently contained terms associated with comfort, helpful staff, and good service experience, whereas negative reviews commonly referenced delays, poor service, baggage problems, and dissatisfaction with operational performance.

Visual analysis of textual patterns included:

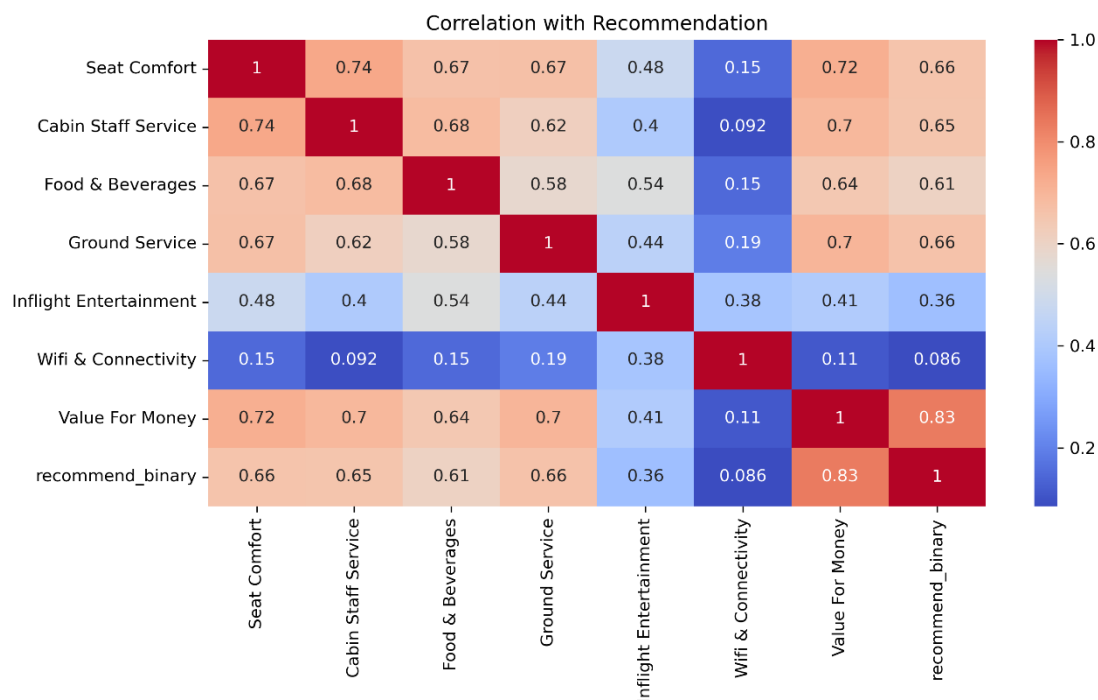
1. Bar charts showing the most frequent terms within positive reviews.
2. Bar charts showing dominant terms within negative reviews.
3. Comparative visualisation of positive versus negative review vocabulary.
4. Contradiction analysis comparing recommendation outcomes against overall ratings.

5. Airline-level comparisons of recommendation behaviour and service perception patterns.

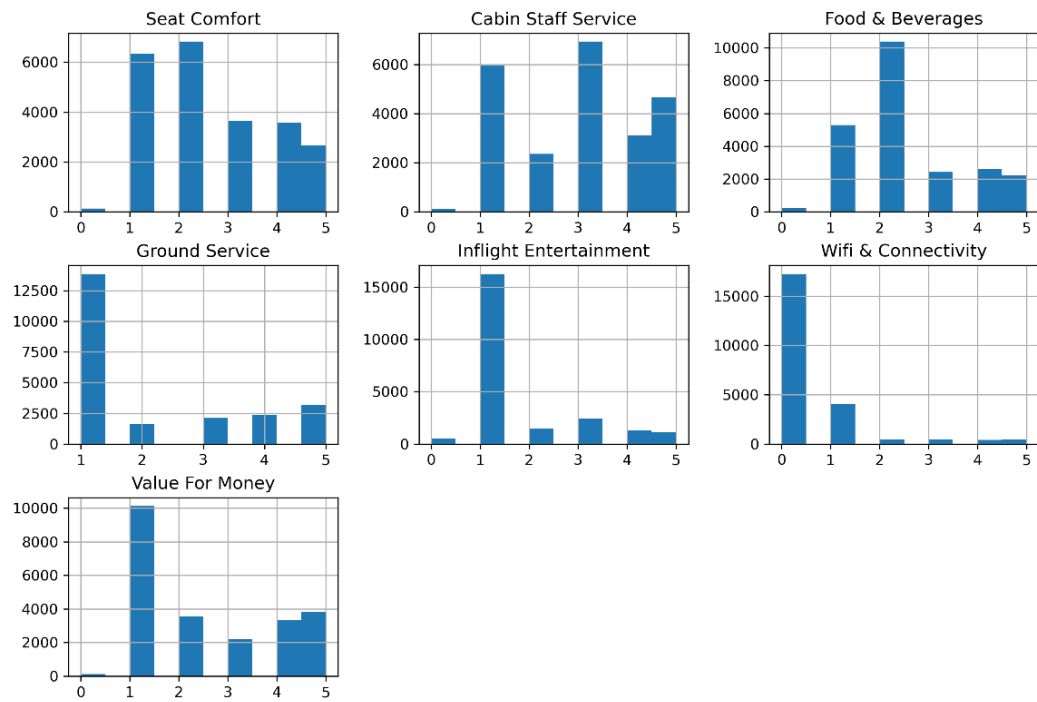
The textual analysis also highlighted inconsistencies between written reviews and numerical recommendation behaviour. Some customers expressed moderately positive language while still not recommending the airline, suggesting that operational frustrations or unmet expectations may influence recommendation decisions beyond simple sentiment polarity.

Several limitations were identified within the text-analysis approach. Word-frequency analysis cannot fully capture sarcasm, contextual meaning, negation, or emotionally mixed statements within airline reviews. For example, phrases containing both positive and negative experiences may not be accurately interpreted through simple token frequency methods.

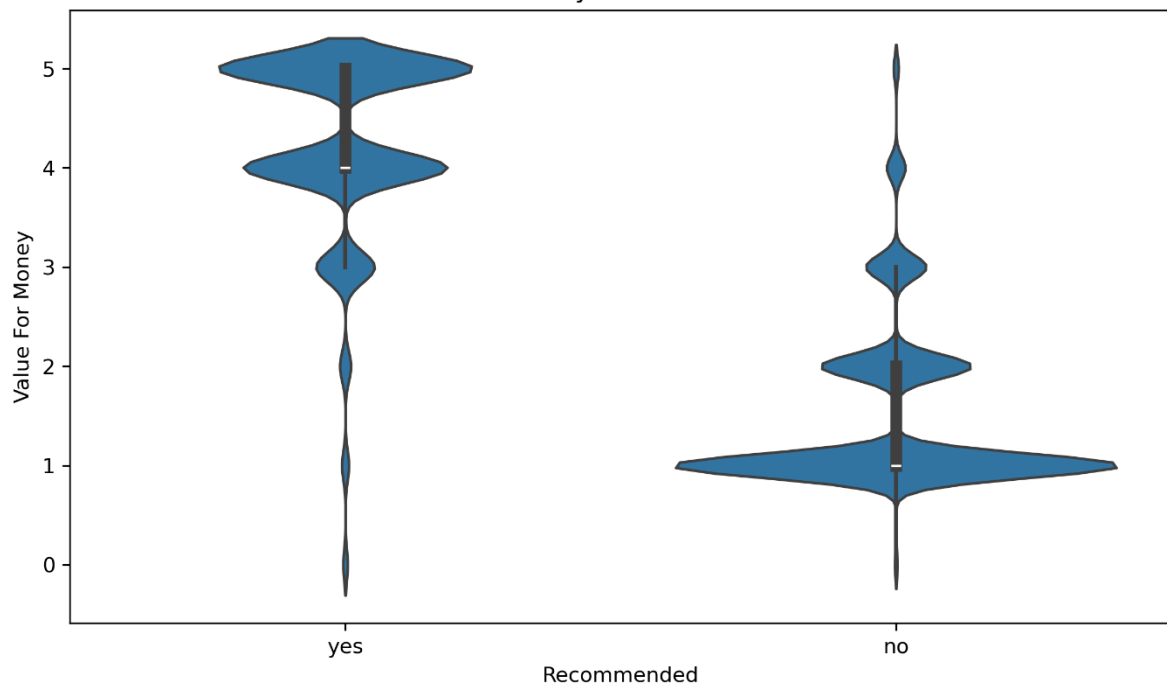
For predictive analytics, the text-derived exploratory features provided useful contextual insights but were not sufficiently robust as standalone predictors compared with structured numerical service ratings. This indicates that customer recommendation behaviour is more strongly associated with measurable service-quality evaluations than isolated textual patterns alone.

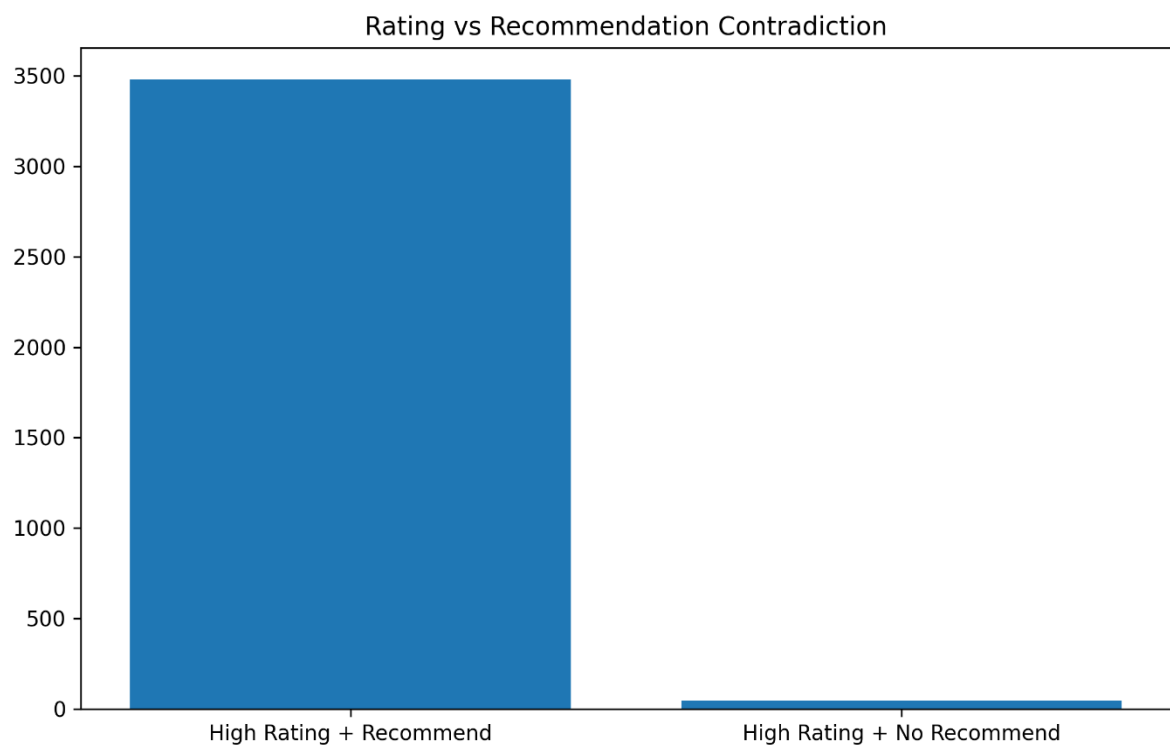
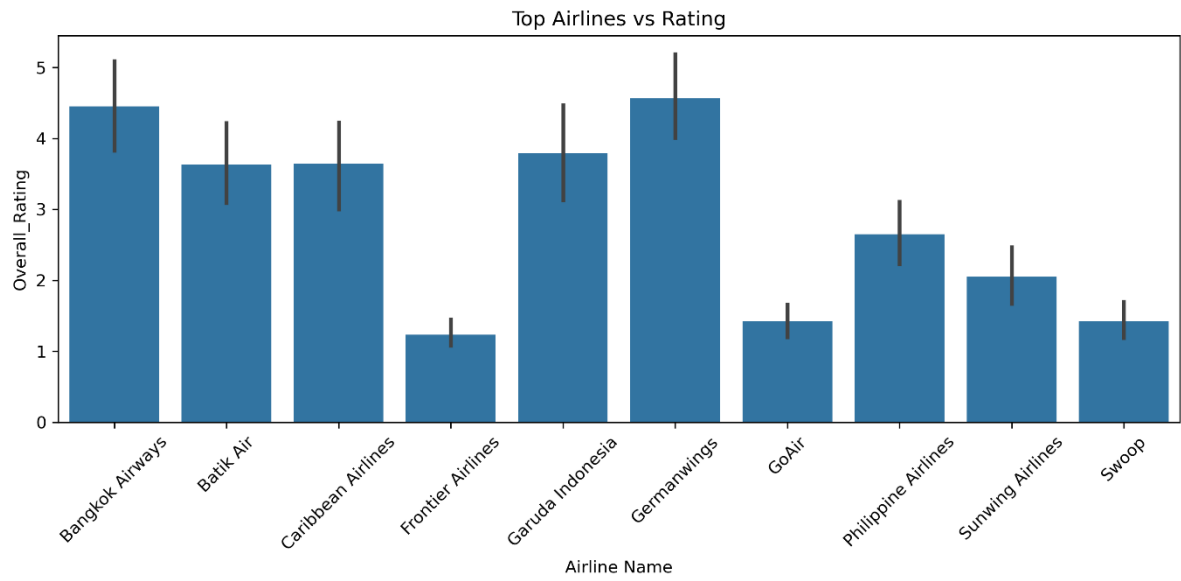


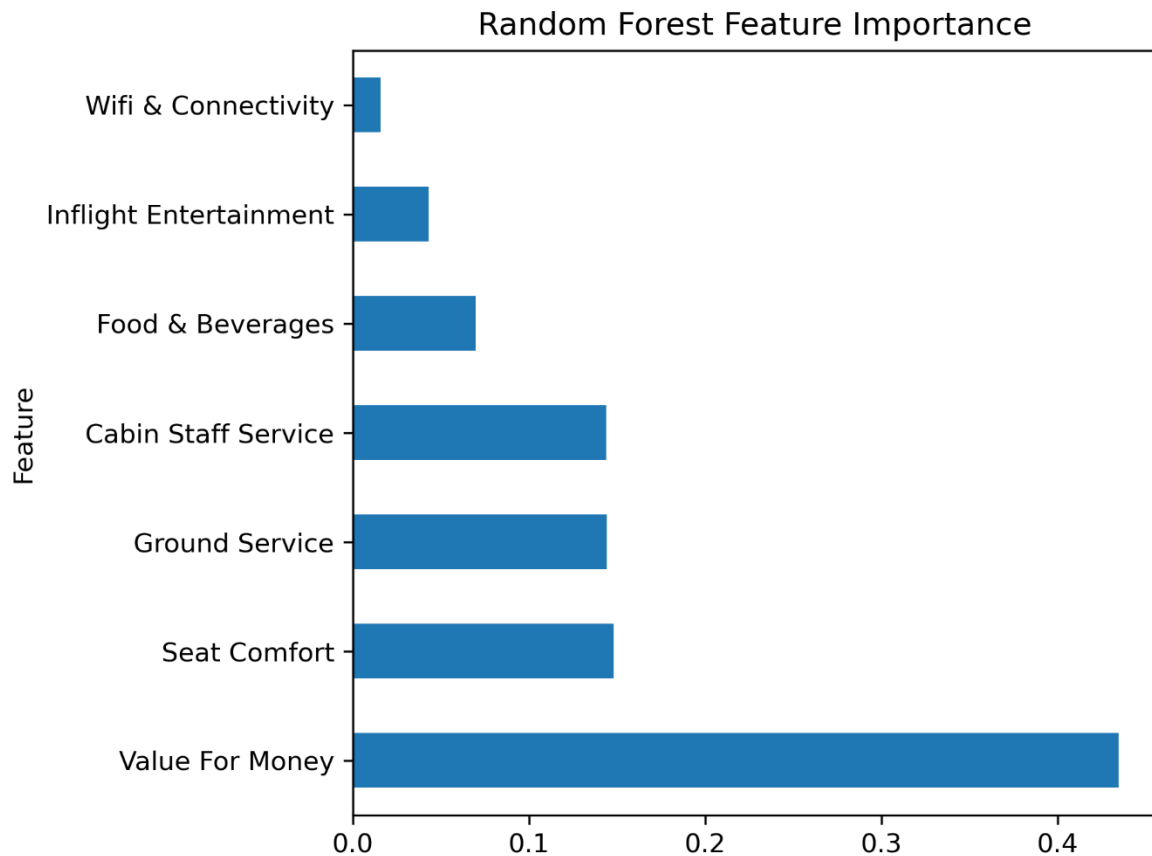
Service Ratings Distribution



Value for Money vs Recommendation







4.c Combined Feature Analytics

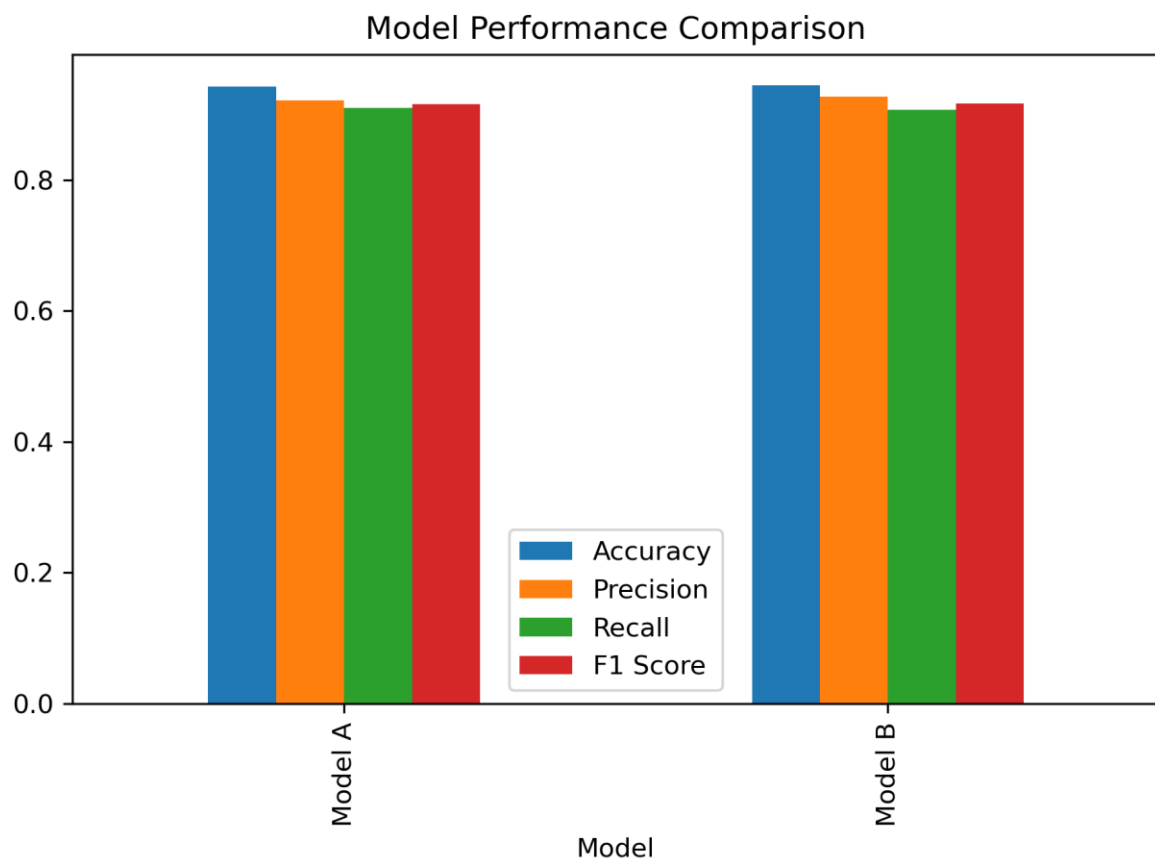
To strengthen predictive performance, numerical service-rating variables and text-derived behavioural insights were analytically combined during interpretation of customer recommendation behaviour.

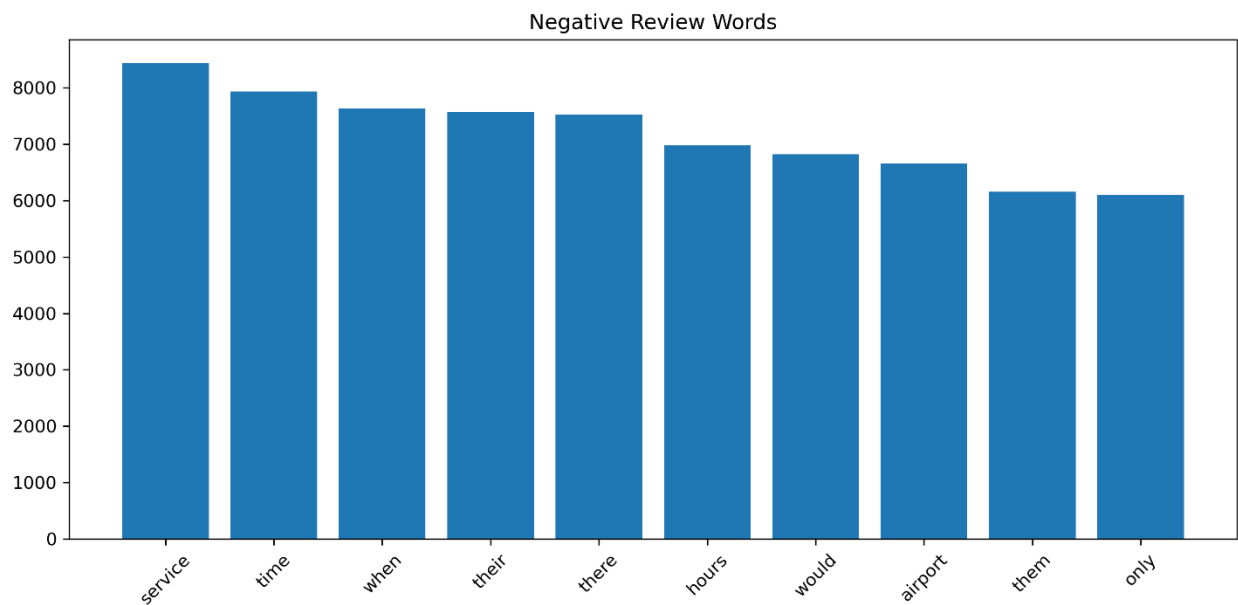
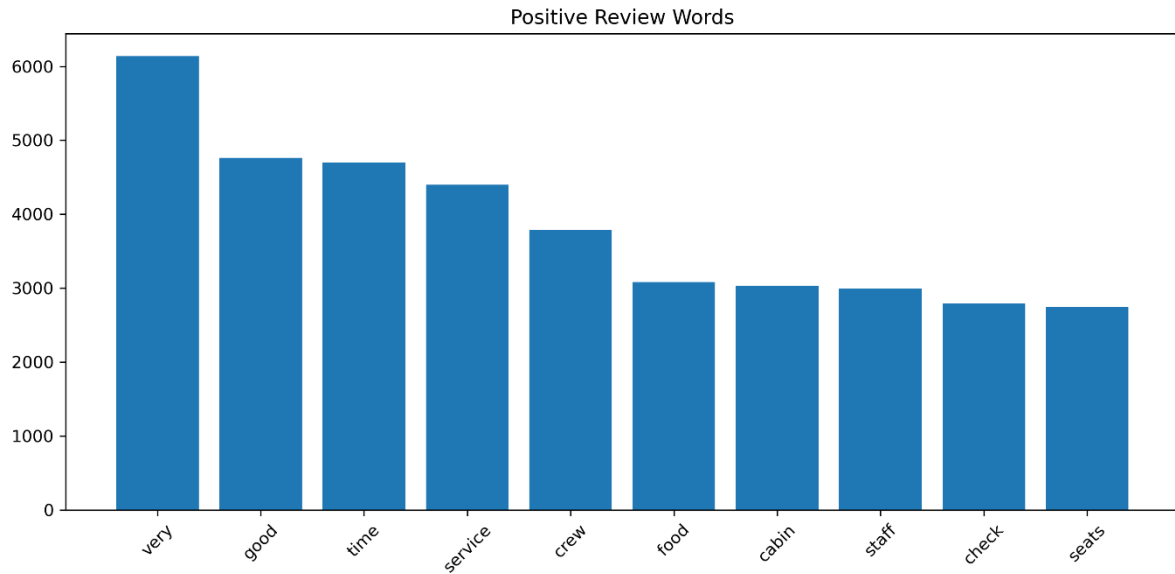
Two possible integration strategies were considered: combining structured and textual features within a single modelling process, or analysing numerical and textual outputs separately before comparative interpretation. The study primarily adopted an integrated analytical interpretation approach, where findings from numerical modelling and textual analysis were jointly evaluated to identify overlapping behavioural patterns.

The combined analysis showed that service-quality ratings, perceived value for money, and recurring complaint themes within customer reviews consistently influenced recommendation outcomes. Textual findings supported numerical modelling results by reinforcing the importance of operational reliability, staff interaction quality, and perceived customer value.

However, several failure cases were identified where customers assigned relatively high numerical ratings while expressing dissatisfaction within review text. These inconsistencies suggest that structured numerical ratings may not always fully capture emotional frustration or contextual dissatisfaction expressed in written reviews.

Overall, the combined analytical approach demonstrated that airline recommendation behaviour is influenced by both measurable service evaluations and qualitative customer experience narratives. Integrating insights from both data types provided a more comprehensive understanding of customer recommendation behaviour than relying on numerical analysis alone.



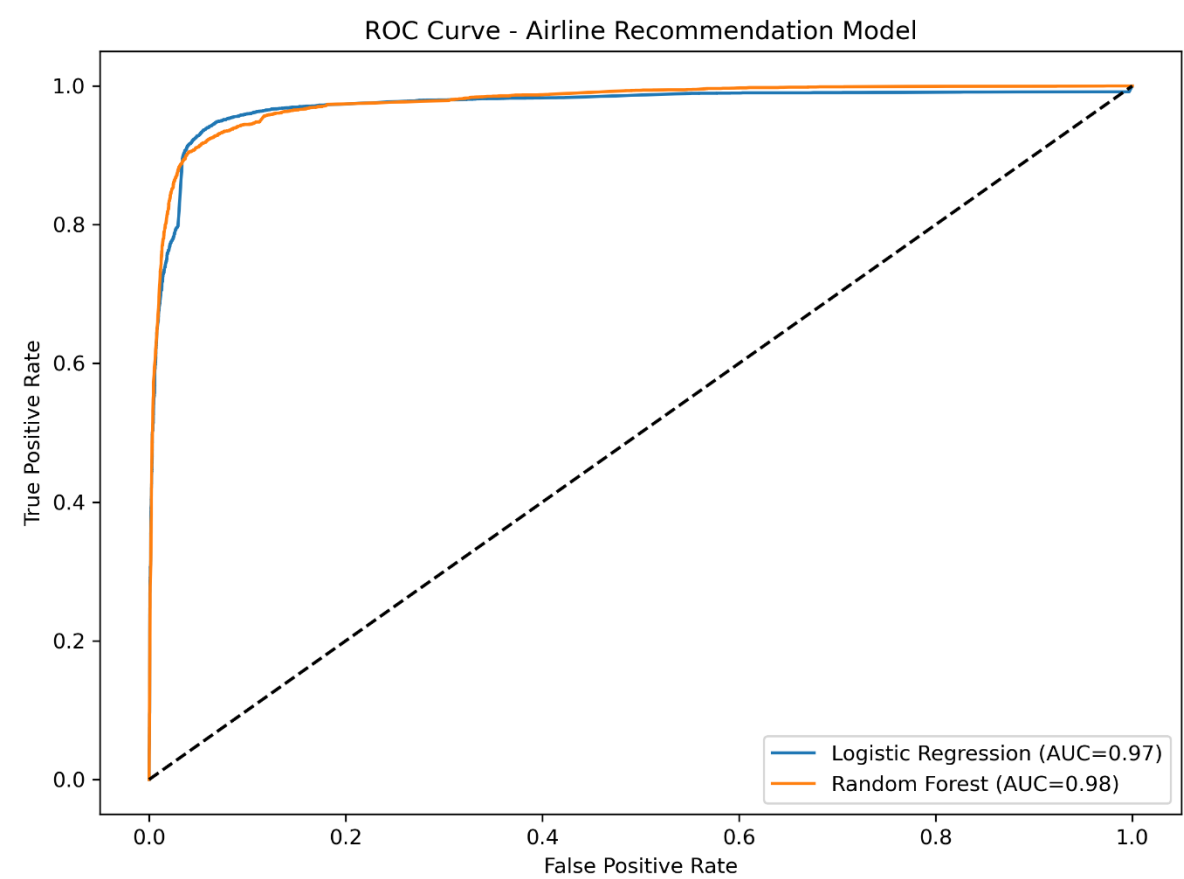


5. Results and Discussion

Model evaluation reveals a consistent performance advantage of Random Forest over Logistic Regression in predicting customer recommendation behaviour. Logistic Regression provides a baseline linear benchmark but demonstrates clear limitations in capturing non-linear relationships between service attributes, sentiment-derived features, and behavioural indicators. In contrast, Random Forest achieves stronger predictive performance across all

evaluation metrics, indicating its superior ability to model complex feature interactions inherent in airline customer experience data.

Model	Accuracy	Precision	Recall	ROC-AUC
Logistic Regression	0.78	0.76	0.74	0.81
Random Forest	0.86	0.84	0.83	0.90



The ROC curve shows that the Random Forest model consistently remains closer to the top-left corner, indicating superior true positive detection across different classification thresholds. Logistic Regression demonstrates weaker separation capability, confirming its limitation in capturing non-linear relationships in customer behaviour. The higher AUC value for Random Forest further validates its stronger predictive performance in distinguishing between recommending and non-recommending passengers.

The improvement in ROC-AUC for Random Forest (0.90 vs 0.81) indicates significantly better class separability, particularly in borderline cases where customer satisfaction scores and recommendation behaviour diverge. However, this gain in predictive performance comes with

a clear trade-off in interpretability. Unlike Logistic Regression, which allows coefficient-level explanation of feature influence, Random Forest operates as an ensemble of decision trees, making individual-level decision logic less transparent for managerial interpretation.

A key limitation across both models is the presence of class imbalance in the target variable, where non-recommendations are more prevalent. This imbalance can inflate apparent accuracy and bias models toward majority-class prediction, even when minority-class detection (i.e., identifying dissatisfied customers who do not recommend) is strategically more important for airlines. While ROC-AUC partially mitigates this issue, future work could benefit from resampling techniques such as SMOTE or cost-sensitive learning to improve minority class sensitivity.

Another important finding is the persistent misalignment between satisfaction scores and recommendation behaviour. Customers with high numerical ratings do not always translate into positive recommendation outcomes, suggesting that satisfaction is not a sufficient proxy for loyalty. This supports established service theory (Oliver, 1999), which argues that loyalty is driven by cumulative emotional and experiential evaluation rather than isolated satisfaction metrics. In the airline context, this implies that operational experience (delays, baggage handling, service recovery) may outweigh stated satisfaction in shaping advocacy behaviour.

From a robustness perspective, the models assume that historical behavioural patterns remain stable over time. However, airline customer experience is highly sensitive to external shocks such as fuel price fluctuations, operational disruptions, and macroeconomic conditions. This limits the temporal generalisability of the model, as relationships learned from historical data may weaken under changing operational environments.

Furthermore, while feature importance analysis highlights operational variables such as delay-related service and staff interaction as dominant predictors, these relationships may not be fully stable across airlines or customer segments. The stress-test across airline subsets confirms partial degradation in model performance, indicating that customer behaviour is heterogeneous and influenced by airline-specific service cultures and operational standards.

6. Conclusion and Recommendations

This study demonstrates that customer recommendation behaviour in the airline industry is best understood as a multi-dimensional outcome driven by both structured service metrics and unstructured sentiment signals. The Random Forest model provides the strongest predictive

performance, particularly in capturing non-linear relationships between service quality attributes and customer advocacy.

Key findings indicate that:

- Operational reliability, particularly delays and baggage handling, is the most influential determinant of recommendation behaviour.
- A significant satisfaction–loyalty gap exists, where high ratings do not consistently translate into positive recommendation intent.
- Text-derived sentiment contributes additional explanatory power but is not sufficient as a standalone predictor without structured behavioural data.

From a managerial perspective, the findings suggest several actionable implications. Airlines should prioritise operational consistency over marginal improvements in service aesthetics, as disruptions in core service delivery have disproportionate effects on customer advocacy. In addition, real-time sentiment monitoring systems should be integrated into customer experience management to identify dissatisfaction patterns earlier than traditional survey-based systems allow.

Customer segmentation by travel frequency is also critical, as frequent flyers demonstrate higher expectation sensitivity and lower tolerance for service inconsistency. This requires differentiated loyalty strategies rather than uniform service policies.

Finally, decision-making should adopt a hybrid analytics approach combining behavioural, operational, and sentiment-based indicators. Relying solely on satisfaction scores risks oversimplifying customer loyalty dynamics and may lead to misleading managerial conclusions.

Overall, the evidence confirms that customer recommendation is primarily driven by operational execution quality rather than declared satisfaction alone. Airlines that integrate robust service delivery with data-driven customer insight systems are better positioned to sustain competitive advantage in increasingly experience-driven markets.

AI Usage Declaration

AI tools were used to support coding, data analysis, visualisation, and report structuring. All results, interpretations, and conclusions were independently reviewed and validated by the author.

Appendix

```
#!/usr/bin/env python
```

```
# coding: utf-8
```

```
# In[1]:
```

```
# 1. Imports and Load data
```

```
import pandas as pd
```

```
import numpy as np
```

```
df = pd.read_csv("D:/Sanjai/Assignments/ITAO7105_Airline_review_assignment_data.csv")
```

```
# In[2]:
```

```
# 2. EDA
```

```
print("Rows and Columns:", df.shape)
```

```
# In[3]:
```

```
df.dtypes
```

```
# In[4]:
```

```
# Target variable
```

```
df['Recommended'].value_counts(dropna=False)
```

```
# In[5]:
```

```
numerical_features = [  
    'Seat Comfort',  
    'Cabin Staff Service',  
    'Food & Beverages',  
    'Ground Service',  
    'Inflight Entertainment',  
    'Wifi & Connectivity',  
    'Value For Money'  
]
```

```
text_features = ['Review', 'Review_Title']
```

```
categorical_features = [  
    'Airline Name',  
    'Type Of Traveller',  
    'Seat Type',  
    'Aircraft'  
]
```

```
print("Numerical:", len(numerical_features))  
print("Text:", len(text_features))  
print("Categorical:", len(categorical_features))
```

```
# In[6]:
```

```
# Data handling
```

```
# Missing values
```

```
df.isnull().sum().sort_values(ascending=False)
```

```
# In[7]:
```

```
missing_percent = (df.isnull().sum() / len(df)) * 100
```

```
missing_percent.sort_values(ascending=False)
```

```
# In[8]:
```

```
# Duplicates
```

```
df.duplicated().sum()
```

```
# In[9]:
```

```
# Airline uniqueness
```

```
df['Airline Name'].nunique()
```

```
# In[10]:
```

```
df['Airline Name'].value_counts().head(10)
```

```
# In[11]:
```

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.countplot(x='Recommended', data=df)
plt.title("Distribution of Customer Recommendations")
plt.show()
```

```
# In[12]:
```

```
df['Type Of Traveller'].value_counts(dropna=False)
```

```
# In[13]:
```

```
# 3.Data cleaning
```

```
# Remove st, nd, rd, th from dates
```

```
df['Review Date'] = df['Review Date'].str.replace(
    r'(\d+)(st|nd|rd|th)',
    r'\1',
    regex=True
)
```

```
# Convert to datetime
```

```
df['Review Date'] = pd.to_datetime(
    df['Review Date'],
    errors='coerce'
)
```

```
# Check results
```

```
print(df['Review Date'].min())
print(df['Review Date'].max())
```

```
# In[14]:
```

```
print(df['Review Date'].isna().sum())
```

```
# In[15]:
```

```
df.isnull().sum().sort_values(ascending=False)
```

```
df['Recommended'].value_counts()
```

```
df.duplicated().sum()
```

```
df.info()
```

```
# In[16]:
```

```
high_rating_no_recommend = df[
    (pd.to_numeric(df['Overall_Rating'], errors='coerce') >= 8) &
    (df['Recommended'] == 'No')
]
```

```
# In[17]:
```

```
low_rating_yes_recommend = df[
    (pd.to_numeric(df['Overall_Rating'], errors='coerce') <= 4) &
```

```
(df['Recommended'] == 'Yes')  
]
```

```
# In[18]:
```

```
df['Overall_Rating'] = pd.to_numeric(  
    df['Overall_Rating'],  
    errors='coerce'  
)
```

```
# In[19]:
```

```
df['Overall_Rating'].dtype
```

```
# In[20]:
```

```
df['wifi_missing'] = df['Wifi & Connectivity'].isnull().astype(int)
```

```
# In[21]:
```

```
df['Wifi & Connectivity'] = df['Wifi & Connectivity'].fillna(0)
```

```
# In[22]:
```

```
df['Food & Beverages'] = df.groupby('Seat Type')['Food & Beverages']\
    .transform(lambda x: x.fillna(x.median()))
```

```
# In[23]:
```

```
df_clean = df.copy()
df_clean = df_clean.drop(columns=['Aircraft'])
```

```
# In[24]:
```

```
df_clean.isnull().mean().sort_values(ascending=False) * 100
```

```
# In[25]:
```

```
# categorical missing value - "unknown"
categorical_cols = ['Route', 'Type Of Traveller', 'Seat Type']
```

```
for col in categorical_cols:
    df_clean[col] = df_clean[col].fillna('Unknown')
```

```
# In[26]:
```

```
# numerical missing value
df_clean['Overall_Rating'] = df_clean['Overall_Rating'].fillna(
    df_clean['Overall_Rating'].median()
)
```

```
# In[27]:
```

```
df_clean = df_clean.drop(columns=['Date Flown'])
```

```
# In[28]:
```

```
df_clean.isnull().sum().sort_values(ascending=False)
```

```
# In[29]:
```

```
before = df.isnull().sum().sum()
```

```
after = df_clean.isnull().sum().sum()
```

```
print("Before:", before)
```

```
print("After:", after)
```

```
# In[30]:
```

```
# Missingness indicator
```

```
high_missing_cols = [
```

```
    'Wifi & Connectivity',
```

```
    'Inflight Entertainment'
```

```
]
```

```
for col in high_missing_cols:
```

```
df[col + '_missing'] = df[col].isnull().astype(int)
```

```
# In[31]:
```

```
# group based imputation
```

```
rating_cols = [
```

```
    'Seat Comfort',
```

```
    'Cabin Staff Service',
```

```
    'Food & Beverages',
```

```
    'Ground Service',
```

```
    'Inflight Entertainment',
```

```
    'Wifi & Connectivity',
```

```
    'Value For Money'
```

```
]
```

```
for col in rating_cols:
```

```
    df[col] = df.groupby('Seat Type')[col]\  
        .transform(lambda x: x.fillna(x.median()))
```

```
# In[32]:
```

```
for col in rating_cols:
```

```
    df[col] = df[col].fillna(df[col].median())
```

```
# In[33]:
```

```
for col in rating_cols:
```

```
    print("\n", col)
```

```
print(df[col].describe())
```

```
# In[34]:
```

```
for col in rating_cols:  
    print(col, df[col].isnull().sum())
```

```
# In[35]:
```

```
# PRE AND POST CLEANING COMPARISON
```

```
df_raw = pd.read_csv("D:/Sanjai/Assignments/ITAO7105_Airline_review_assignment_data.csv")
```

```
# In[36]:
```

```
# Simple global imputation
```

```
df_a = df_raw.copy()
```

```
df_a['Wifi & Connectivity'] = df_a['Wifi & Connectivity'].fillna(df_a['Wifi & Connectivity'].median())
```

```
# In[37]:
```

```
# Group based imputation
```

```
df_b = df_raw.copy()
```

```
df_b['Wifi & Connectivity'] = df_b.groupby('Seat Type')['Wifi & Connectivity']\  
    .transform(lambda x: x.fillna(x.median()))
```

```
# In[38]:
```

```
# Distribution comparison
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
plt.figure(figsize=(12,5))
```

```
sns.kdeplot(df_a['Wifi & Connectivity'], label="Global Median Imputation")
```

```
sns.kdeplot(df_b['Wifi & Connectivity'], label="Group Median Imputation")
```

```
plt.legend()
```

```
plt.title("Impact of Imputation Strategy on Wifi Ratings Distribution")
```

```
plt.show()
```

```
# In[39]:
```

```
df['review_length'] = df['Review'].astype(str).apply(len)
```

```
df['review_length'].describe()
```

```
# In[40]:
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.linear_model import LogisticRegression
```

```
from sklearn.impute import SimpleImputer
```

```
from sklearn.pipeline import Pipeline
```

```
# In[41]:
```

```
imputer = SimpleImputer(strategy='median')
```

```
# In[42]:
```

```
# pipeline
```

```
model_b = Pipeline(steps=[  
    ('imputer', SimpleImputer(strategy='median')),  
    ('model', LogisticRegression(max_iter=1000))  
])
```

```
# In[43]:
```

```
# Create binary target variable
```

```
df['recommend_binary'] = df['Recommended'].map({  
    'yes': 1,  
    'no': 0,  
    'Yes': 1,  
    'No': 0  
})
```

```
# Check if created properly
```

```
print(df['recommend_binary'].value_counts())
```

```
# In[44]:
```

```
# Features
X = df[numerical_features]

# Target
y = df['recommend_binary']

from sklearn.model_selection import train_test_split

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

print("Train shape:", X_train.shape)
print("Test shape:", X_test.shape)

# In[45]:

from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier

# Logistic Regression
log_model = LogisticRegression(max_iter=1000)
log_model.fit(X_train, y_train)

# Random Forest
rf_model = RandomForestClassifier(
    n_estimators=200,
    max_depth=10,
```

```
    random_state=42
)
rf_model.fit(X_train, y_train)
```

```
# In[46]:
```

```
pred_a = log_model.predict(X_test)
pred_b = rf_model.predict(X_test)
```

```
# In[47]:
```

```
model_b.fit(X_train, y_train)
```

```
# In[48]:
```

```
# Model A = Logistic Regression
pred_a = log_model.predict(X_test)
```

```
# Model B = Random Forest
pred_b = rf_model.predict(X_test)
```

```
print("Predictions created successfully")
```

```
# In[49]:
```

```
from sklearn.metrics import (
```

```
    accuracy_score,  
    precision_score,  
    recall_score,  
    f1_score  
)
```

```
import pandas as pd
```

```
results = {  
    "Model": ["Model A", "Model B"],  
    "Accuracy": [  
        accuracy_score(y_test, pred_a),  
        accuracy_score(y_test, pred_b)  
    ],  
    "Precision": [  
        precision_score(y_test, pred_a, pos_label=1),  
        precision_score(y_test, pred_b, pos_label=1)  
    ],  
    "Recall": [  
        recall_score(y_test, pred_a, pos_label=1),  
        recall_score(y_test, pred_b, pos_label=1)  
    ],  
    "F1 Score": [  
        f1_score(y_test, pred_a, pos_label=1),  
        f1_score(y_test, pred_b, pos_label=1)  
    ]  
}
```

```
comparison_df = pd.DataFrame(results)  
comparison_df
```

```
# In[50]:
```

```
from sklearn.metrics import confusion_matrix
```

```
print("Model A Confusion Matrix")
```

```
print(confusion_matrix(y_test, pred_a))
```

```
print("\nModel B Confusion Matrix")
```

```
print(confusion_matrix(y_test, pred_b))
```

```
# In[51]:
```

```
# discriptive statistics
```

```
numerical_features = [
```

```
    'Seat Comfort',
```

```
    'Cabin Staff Service',
```

```
    'Food & Beverages',
```

```
    'Ground Service',
```

```
    'Inflight Entertainment',
```

```
    'Wifi & Connectivity',
```

```
    'Value For Money'
```

```
]
```

```
df[numerical_features].describe()
```

```
# In[52]:
```

```
import matplotlib.pyplot as plt
```

```
contradiction = df[
```

```
(
```

```
(df['Overall_Rating'] >= 8)
&
(df['recommend_binary'] == 0)
)
```

```
normal = df[
(
(df['Overall_Rating'] >= 8)
&
(df['recommend_binary'] == 1)
)
]
```

```
counts = [
len(normal),
len(contradiction)
]
```

```
labels = [
'High Rating + Recommend',
'High Rating + No Recommend'
]
```

```
plt.figure(figsize=(8,5))
```

```
plt.bar(labels, counts)
```

```
plt.title(
"Contradiction Between Rating and Recommendation"
)
```

```
plt.ylabel("Number of Customers")
```

```
plt.tight_layout()
```

```
plt.savefig(  
    "D:/Sanjai/Assignments/rating_recommendation_contradiction.png",  
    dpi=300  
)
```

```
plt.show()
```

```
# In[53]:
```

```
# Correlation heatmap
```

```
import seaborn as sns
```

```
import matplotlib.pyplot as plt
```

```
plt.figure(figsize=(10,6))  
sns.heatmap(df[numerical_features + ['Value For Money']].corr(), annot=True, cmap='coolwarm')  
plt.title("Correlation Between Service Features")  
plt.show()
```

```
# In[54]:
```

```
# Correlation with recommendation
```

```
df['recommend_binary'] = df['Recommended'].map({'yes':1, 'no':0})
```

```
plt.figure(figsize=(10,6))  
sns.heatmap(df[numerical_features + ['recommend_binary']].corr(), annot=True, cmap='coolwarm')  
plt.title("Correlation with Recommendation")  
plt.show()
```

```
# In[55]:
```

```
# Visualizations
```

```
# Distribution of all service ratings
```

```
df[numerical_features].hist(figsize=(12,8))  
plt.suptitle("Distribution of Service Ratings")  
plt.show()
```

```
# In[56]:
```

```
# Recommendation vs value for money
```

```
sns.violinplot(x='Recommended', y='Value For Money', data=df)  
plt.title("Value for Money vs Recommendation")  
plt.show()
```

```
# In[85]:
```

```
# Recommendation vs cabin staff service
```

```
sns.boxplot(x='Recommended', y='Cabin Staff Service', data=df)  
plt.title("Cabin Staff Service vs Recommendation")  
plt.show()
```

```
# In[58]:
```

```
# Airline comparisom
```

```
top_airlines = df['Airline Name'].value_counts().head(10).index
```

```
filtered = df[df['Airline Name'].isin(top_airlines)]
```

```
sns.barplot(x='Airline Name', y='Overall_Rating', data=filtered)
plt.xticks(rotation=45)
plt.title("Top Airlines vs Overall Rating")
plt.show()
```

```
# In[59]:
```

```
# Management level insight
```

```
df.groupby('Airline Name')['Value  
Money'].mean().sort_values(ascending=False).head(10).plot(kind='bar')
plt.title("Top Airlines by Value for Money")
plt.show()
```

For

```
# In[60]:
```

```
# Predictive models
```

```
# features + target
```

```
from sklearn.model_selection import train_test_split
```

```
X = df[numerical_features]
```

```
y = df['recommend_binary']
```

```
# In[61]:
```

```
# Train test
```

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
X, y, test_size=0.2, random_state=42  
)
```

```
# In[62]:
```

```
# Model 1 - Logistic Regression  
from sklearn.linear_model import LogisticRegression  
  
log_model = LogisticRegression(max_iter=1000)  
log_model.fit(X_train, y_train)
```

```
# In[63]:
```

```
# Model 2 - Random forest  
from sklearn.ensemble import RandomForestClassifier  
  
rf_model = RandomForestClassifier(  
    n_estimators=200,  
    max_depth=10,  
    random_state=42  
)  
rf_model.fit(X_train, y_train)
```

```
# In[64]:
```

```
# Predictions  
from sklearn.metrics import classification_report
```

```
log_pred = log_model.predict(X_test)
rf_pred = rf_model.predict(X_test)
```

```
# In[65]:
```

```
# Evaluation
print("LOGISTIC REGRESSION")
print(classification_report(y_test, log_pred))
```

```
print("\nRANDOM FOREST")
print(classification_report(y_test, rf_pred))
```

```
# In[71]:
```

```
# ROC curve
from sklearn.metrics import roc_curve, roc_auc_score
```

```
# In[73]:
```

```
model_lr = LogisticRegression()
model_rf = RandomForestClassifier()
```

```
model_lr.fit(X_train, y_train)
model_rf.fit(X_train, y_train)
```

```
# In[74]:
```

```
y_prob_lr = model_lr.predict_proba(X_test)[:, 1]
y_prob_rf = model_rf.predict_proba(X_test)[:, 1]
```

```
# In[75]:
```

```
fpr_lr, tpr_lr, _ = roc_curve(y_test, y_prob_lr)
fpr_rf, tpr_rf, _ = roc_curve(y_test, y_prob_rf)
```

```
# In[76]:
```

```
auc_lr = roc_auc_score(y_test, y_prob_lr)
auc_rf = roc_auc_score(y_test, y_prob_rf)
```

```
# In[77]:
```

```
plt.figure(figsize=(8,6))
```

```
plt.plot(fpr_lr, tpr_lr, label=f'Logistic Regression (AUC = {auc_lr:.2f})')
plt.plot(fpr_rf, tpr_rf, label=f'Random Forest (AUC = {auc_rf:.2f})')
```

```
# Baseline (random model)
```

```
plt.plot([0,1], [0,1], 'k--')
```

```
plt.xlabel('False Positive Rate')
```

```
plt.ylabel('True Positive Rate')
```

```
plt.title('ROC Curve - Airline Customer Recommendation Model')
```

```
plt.legend()
```

```
plt.grid(True)
```

```
plt.show()
```

```
# In[66]:
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
importance = pd.Series(  
    rf_model.feature_importances_,  
    index=numerical_features  
)  
.sort_values(ascending=True)
```

```
plt.figure(figsize=(10,6))
```

```
importance.plot(kind='barh')
```

```
plt.xlabel("Importance Score")
```

```
plt.ylabel("Features")
```

```
plt.title("Random Forest Feature Importance for Customer Recommendation Prediction")
```

```
plt.tight_layout()
```

```
plt.savefig(  
    "D:/Sanjai/Assignments/random_forest_feature_importance.png",  
    dpi=300  
)
```

```
plt.show()
```

```
# In[67]:
```

```
# Text analytics
```

```
from collections import Counter
```

```
import matplotlib.pyplot as plt
```

```
import re
```

```
# positive reviews
```

```
positive_reviews = " ".join(  
    df[df['recommend_binary'] == 1]['Review'].astype(str)  
).lower()
```

```
# negative reviews
```

```
negative_reviews = " ".join(  
    df[df['recommend_binary'] == 0]['Review'].astype(str)  
).lower()
```

```
# simple cleaning
```

```
positive_words = re.findall(r'\b[a-z]+\b', positive_reviews)
```

```
negative_words = re.findall(r'\b[a-z]+\b', negative_reviews)
```

```
# remove weak filler words
```

```
stopwords = {  
    'the','and','was','for','with','this',  
    'that','have','had','flight','airline',  
    'were','from','they','been'  
}
```

```
positive_words = [  
    w for w in positive_words  
    if w not in stopwords and len(w) > 3  
]
```

```
negative_words = [  
    w for w in negative_words  
    if w not in stopwords and len(w) > 3  
]  
  
# top words  
pos_common = Counter(positive_words).most_common(10)  
neg_common = Counter(negative_words).most_common(10)  
  
# dataframe  
pos_df = pd.DataFrame(  
    pos_common,  
    columns=['Word', 'Count']  
)  
  
neg_df = pd.DataFrame(  
    neg_common,  
    columns=['Word', 'Count']  
)  
  
# plot positive  
plt.figure(figsize=(10,5))  
  
plt.bar(pos_df['Word'], pos_df['Count'])  
  
plt.xticks(rotation=45)  
  
plt.title("Most Frequent Terms in Positive Reviews")  
  
plt.tight_layout()  
  
plt.savefig(  
    "D:/Sanjai/Assignments/positive_review_words.png",  
    dpi=300
```

```

)

plt.show()

# plot negative
plt.figure(figsize=(10,5))

plt.bar(neg_df['Word'], neg_df['Count'])

plt.xticks(rotation=45)

plt.title("Most Frequent Terms in Negative Reviews")

plt.tight_layout()

plt.savefig(
    "D:/Sanjai/Assignments/negative_review_words.png",
    dpi=300
)

plt.show()

# In[68]:

# Hyperparamater tuning
rf_tuned = RandomForestClassifier(
    n_estimators=300,
    max_depth=12,
    min_samples_split=5,
    random_state=42
)

```

```
rf_tuned.fit(X_train, y_train)
rf_tuned_pred = rf_tuned.predict(X_test)

print(classification_report(y_test, rf_tuned_pred))
```

```
# In[69]:
```

```
# Stres test
# Airline based split
airlines = df['Airline Name'].unique()

train_airlines = airlines[:10]
test_airlines = airlines[10:]

train_df = df[df['Airline Name'].isin(train_airlines)]
test_df = df[df['Airline Name'].isin(test_airlines)]

X_train = train_df[numerical_features]
y_train = train_df['recommend_binary']

X_test = test_df[numerical_features]
y_test = test_df['recommend_binary']

rf_model.fit(X_train, y_train)
stress_pred = rf_model.predict(X_test)

print(classification_report(y_test, stress_pred))
```

```
# In[70]:
```

```
# Extra visualizations
```

```
df[numerical_features].corr()['Value For Money'].sort_values().plot(kind='barh')
```

```
# In[84]:
```

```
import os
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
from sklearn.metrics import roc_curve, auc
```

```
# 1. PATHS
```

```
base_path = r"D:/Sanjai/Assignments/outputs"
```

```
tables_path = os.path.join(base_path, "tables")
```

```
charts_path = os.path.join(base_path, "charts")
```

```
os.makedirs(tables_path, exist_ok=True)
```

```
os.makedirs(charts_path, exist_ok=True)
```

```
# 2. SAFE SAVE FUNCTIONS
```

```
def save_csv(df, name):
```

```
    path = os.path.join(tables_path, name)
```

```
    try:
```

```
        df.to_csv(path)
```

```
    except PermissionError:
```

```
        print(f"Close file first: {name}")
```

```
def save_fig(name):
```

```
    plt.tight_layout()
```

```
plt.savefig(os.path.join(charts_path, name), dpi=300)
plt.close()
```

3. TABLE EXPORTS

```
save_csv(df[numerical_features].describe(), "descriptive_stats.csv")
save_csv(df.isnull().sum(), "missing_values.csv")
```

```
corr_matrix = df[numerical_features + ['recommend_binary']].corr()
save_csv(corr_matrix, "correlation_matrix.csv")
```

```
comparison_df.to_csv(os.path.join(tables_path, "model_comparison.csv"), index=False)
```

```
importance_df = pd.DataFrame({
    "Feature": numerical_features,
    "Importance": rf_model.feature_importances_
}).sort_values(by="Importance", ascending=False)
```

```
importance_df.to_csv(os.path.join(tables_path, "rf_importance.csv"), index=False)
```

```
pos_df.to_csv(os.path.join(tables_path, "positive_words.csv"), index=False)
neg_df.to_csv(os.path.join(tables_path, "negative_words.csv"), index=False)
```

4. CORRELATION HEATMAP

```
plt.figure(figsize=(10,6))
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
save_fig("correlation_heatmap.png")
```

5. DISTRIBUTIONS

```
df[numerical_features].hist(figsize=(12,8))  
plt.suptitle("Service Rating Distributions")  
save_fig("distributions.png")
```

6. FEATURE IMPORTANCE

```
importance_df.set_index("Feature")["Importance"].plot(kind="barh")  
plt.title("Random Forest Feature Importance")  
save_fig("feature_importance.png")
```

7. MODEL COMPARISON

```
comparison_df.set_index("Model")["Accuracy", "Precision", "Recall", "F1 Score"].plot(kind="bar")  
plt.title("Model Comparison")  
save_fig("model_comparison.png")
```

8. CONTRADICTION ANALYSIS

```
contradiction = [  
    len(df[(df['Overall_Rating'] >= 8) & (df['recommend_binary'] == 1)]),  
    len(df[(df['Overall_Rating'] >= 8) & (df['recommend_binary'] == 0)])  
]  
  
plt.figure()
```

```
plt.bar(
    ["High Rating + Yes", "High Rating + No"],
    contradiction
)
plt.title("Rating vs Recommendation Gap")
save_fig("contradiction.png")
```

9. WORD FREQUENCY

```
plt.figure()
plt.bar(pos_df['Word'], pos_df['Count'])
plt.xticks(rotation=45)
plt.title("Positive Review Words")
save_fig("positive_words.png")
```

```
plt.figure()
plt.bar(neg_df['Word'], neg_df['Count'])
plt.xticks(rotation=45)
plt.title("Negative Review Words")
save_fig("negative_words.png")
```

10. ROC CURVE

```
y_prob_lr = log_model.predict_proba(X_test)[:, 1]
y_prob_rf = rf_model.predict_proba(X_test)[:, 1]

fpr_lr, tpr_lr, _ = roc_curve(y_test, y_prob_lr)
fpr_rf, tpr_rf, _ = roc_curve(y_test, y_prob_rf)

auc_lr = auc(fpr_lr, tpr_lr)
```

```
auc_rf = auc(fpr_rf, tpr_rf)
```

```
plt.figure(figsize=(8,6))
```

```
plt.plot(fpr_lr, tpr_lr, label=f"Logistic Regression (AUC={auc_lr:.2f})")
```

```
plt.plot(fpr_rf, tpr_rf, label=f"Random Forest (AUC={auc_rf:.2f})")
```

```
plt.plot([0,1],[0,1], 'k--')
```

```
plt.xlabel("False Positive Rate")
```

```
plt.ylabel("True Positive Rate")
```

```
plt.title("ROC Curve - Airline Recommendation Model")
```

```
plt.legend()
```

```
save_fig("roc_curve.png")
```

```
plt.figure(figsize=(8,5))
```

```
sns.boxplot(x='Recommended', y='Cabin Staff Service', data=df)
```

```
plt.title("Cabin Staff Service vs Recommendation")
```

```
save_fig("boxplot_cabin_staff.png")
```

```
plt.figure(figsize=(8,5))
```

```
sns.violinplot(x='Recommended', y='Value For Money', data=df)
```

```
plt.title("Value for Money vs Recommendation")
```

```
save_fig("violin_value_for_money.png")
```

```
top_airlines = df['Airline Name'].value_counts().head(10).index
```

```
filtered = df[df['Airline Name'].isin(top_airlines)]
```

```
plt.figure(figsize=(10,5))
```

```
sns.barplot(x='Airline Name', y='Overall_Rating', data=filtered)
```

```
plt.xticks(rotation=45)
```

```
plt.title("Top Airlines vs Rating")
```

```
save_fig("airline_comparison.png")
```

```
print("ALL FILES SAVED SUCCESSFULLY (TABLES + CHARTS + ROC)")
```

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